EcoEcon: Ecological Resilience Indicators for Systemic Risk—A Retrospective Analysis of the 2008 Financial Crisis with a 2020 Negative Control

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Abstract

Can ecological collapse theory provide early warning indicators for financial crises? We model the US economy as a biological ecosystem—sectors as species, supply chains as mutualistic dependencies, and bankruptcy cascades as extinction events—and apply three bodies of ecological theory to BEA Input-Output and FRED data: May's (1972) stability theorem, Scheffer et al.'s (2009) critical slowing down, and network robustness metrics. We construct an Economic Biodiversity Index (EBI), track network spectral gap, and compute critical slowing down signals across two episodes: the 2008 Global Financial Crisis (endogenous fold bifurcation, signal expected) and the 2020 COVID shock (exogenous perturbation, no signal expected by theory). In a retrospective analysis, two of five ecological signals behaved as theorized before 2008: spectral gap declined monotonically 2002–2007 (Kendall $\tau = -0.87$) and housing starts variance peaked approximately 18 months before crisis onset. Logistic regression trained on these features achieves leave-one-out AUC = 1.000 ($p = 0.034$, permutation test) with zero false positives in cross-validation. The 2020 negative control holds: models trained on 2008 assign near-zero crisis probability to 2015–2019, and four of five ecological signals show no pre-crisis buildup before the exogenous COVID shock, consistent with the theoretical prediction. Spectral gap produces zero false alarms across ten calm-period years (2010–2019). This analysis is explicitly retrospective and limited to one positive crisis episode; generalizability requires multi-crisis validation. We propose ecological resilience indicators as a theoretically grounded and empirically promising direction for systemic risk research.

1 Introduction

Financial crises remain notoriously difficult to predict. Standard early warning systems rely on macroeconomic aggregates—credit growth, asset prices, current account deficits—yet the 2008 Global Financial Crisis exposed the limitations of these approaches: most official forecasts in 2007 projected continued growth [8].

Ecology faces a structurally analogous problem: detecting when a stable ecosystem is approaching collapse. Over five decades, ecologists have developed a robust theoretical framework for understanding stability and collapse in complex systems. **(author?)** [1] proved that complexity itself can be destabilizing—above a critical threshold of diversity and connectance, large random systems become inherently unstable. **(author?)** [2] demonstrated that many complex systems exhibit "critical slowing down" before tipping points: rising variance and autocorrelation in key state variables as the system loses resilience. **(author?)** [3] showed that financial networks exhibit phase transitions where diversification reduces individual risk up to a point, beyond which it amplifies systemic fragility through contagion channels. **(author?)** [4] applied this ecological analogy directly to banking systems, demonstrating that the same complexity-fragility tradeoff governs interbank contagion networks.

This paper operationalizes the ecology-economy analogy by constructing ecological metrics from economic data and testing whether they were elevated before the 2008 crisis. Critically, we also test a theoretically motivated *negative control*: the 2020 COVID shock was an exogenous perturbation rather than an endogenous approach to a bifurcation, so ecological theory predicts that critical slowing down signals should *not* appear before 2020. This contrast—signal before an endogenous crisis, no signal before an exogenous shock—provides stronger evidence than a single positive episode alone.

We make no claim that these signals would have been detectable in real time or that they generalize beyond this single positive episode. That question requires multi-crisis validation outside the scope of this retrospective study.

2 Methods

2.1 Variable Mapping

Table 1 presents the ecological-to-economic translation underlying our metrics. We note one important theoretical correction relative to standard ecological analogies: supplier-buyer relationships in IO networks are structurally closer to *mutualistic* interactions (+/+) than predator-prey interactions (+/−), since both parties benefit from the transaction. This distinction matters because **(author?)** [5] showed that mutualistic networks destabilize more rapidly with increasing diversity and connectance than predator-prey networks—placing economic IO networks in a potentially more fragile stability regime than the predator-prey framing would imply.

Table 1: Ecological-to-economic variable mapping

Ecological Concept	**Economic Analog**
Species	Sector (68 BEA industries)
Mutualistic dependency	Supplier-buyer relationship
Biomass flow	Revenue flow between sectors
Keystone species	Systemically important sector
Extinction cascade	Bankruptcy contagion
Ecosystem biodiversity	Sector centrality distribution (EBI)
Critical slowing down	Rising variance + autocorrelation
Spectral gap collapse	Network fragmentation

2.2 Data Sources

BEA Input-Output Tables. Annual sector-level flows for 68 industries from the Bureau of Economic Analysis (TableID 259), covering two episodes: 2002–2009 (2008 GFC) and 2015–2021 (2020 COVID negative control). Edge weights are normalized to technical coefficients (column-normalized input shares) rather than raw dollar flows, removing nominal GDP and inflation effects from network metrics. The 2001 dot-com episode was excluded due to incompatible industry classification systems between pre-2002 SIC-based and post-2002 NAICS-based BEA IO tables.

FRED Financial Time Series. Monthly data from the Federal Reserve Economic Database, 1998–2022: housing starts (HOUST), bank credit (TOTBKCR), TED spread (TEDRATE, substituting for BAMLH0A0HYM2 which returned empty via API), consumer credit (TOTALSL), and the Federal Funds rate (FEDFUNDS). All rolling windows are strictly trailing (no centering); detrending uses first-differencing.

Crisis Dating. NBER recession dates for both episodes. 2008 episode: December 2007 onset (NBER), June 2009 trough. 2020 episode: March 2020 onset, December 2020 trough. Episode types pre-registered before analysis: 2008 as endogenous fold bifurcation (signal expected), 2020 as exogenous shock (no signal expected).

2.3 Economic Biodiversity Index (EBI)

For each year, we construct a directed, weighted graph $G = (V, E, W)$ where nodes are the 68 BEA sectors and edge weights are technical coefficients. We compute betweenness centrality for each node and form a probability distribution:

$$p_i = \frac{c_i}{\sum_{j=1}^{n} c_j}, \quad i = 1, \ldots, n \tag{1}$$

where c_i is the betweenness centrality of sector i. The Shannon entropy of this distribution is:

$$H = -\sum_{i=1}^{n} p_i \ln p_i \tag{2}$$

The Economic Biodiversity Index is:

$$\mathrm{EBI} = H \times (1 - \mathrm{top3\_share}) \tag{3}$$

where top3_share is the fraction of total centrality held by the three most central sectors. This penalizes apparent diversity by keystone concentration. The choice of betweenness centrality and the top-3 threshold are exploratory; robustness to eigenvector centrality and DebtRank is reserved for future work.

2.4 Spectral Gap

The spectral gap is defined as:

$$\Delta\lambda = \lambda_1 - \lambda_2 \tag{4}$$

where λ_1, λ_2 are the largest and second-largest eigenvalues (by magnitude) of the technical-coefficient adjacency matrix. A collapsing spectral gap indicates network fragmentation—connectivity is becoming dominated by a single mode, reducing redundant pathways.

We note explicitly that this is a *network connectivity property* and is distinct from May's (1972) stability criterion, which concerns the leading eigenvalue of the community (Jacobian) matrix of population dynamics near equilibrium. Computing May's criterion properly from the IO technical-coefficient matrix requires specifying sector adjustment dynamics and self-regulation terms; this is reserved for future work.

2.5 Critical Slowing Down

Following **(author?)** [2], we compute two metrics on monthly FRED series using 24-month strictly trailing rolling windows:

$$\text{Variance}_t = \text{Var}(x_{t-23:t}) \tag{5}$$

$$\text{AR}(1)_t = \text{Corr}(x_{t-23:t-1}, x_{t-22:t}) \tag{6}$$

Rising variance and rising autocorrelation jointly indicate critical slowing down. This is theoretically grounded for fold bifurcations [2]: as a system approaches a tipping point, the dominant eigenvalue of the linearized recovery dynamics approaches zero, causing perturbations to decay increasingly slowly. We note that CSD is only theoretically expected for *endogenous* approaches to bifurcation, not for exogenous shocks—motivating the 2020 negative control.

2.6 Trend Significance

We compute Kendall's τ rank correlation between year and each signal over the pre-crisis window for both episodes. Kendall's τ is appropriate for small samples with potential non-normality and is the standard in the EWS literature [2]. A positive τ indicates a rising trend; the sign is adjusted so that positive always means "moving toward risk" (rising for variance signals, declining for biodiversity/connectivity signals).

2.7 Machine Learning Evaluation

Labels. For the 2008 episode, crisis labels are assigned to 2008–2009 only. The year 2007 is held out as an "advance warning test year"—it receives no label and is tested separately to assess whether models fire before the official crisis onset.

Scaler. A RobustScaler is fitted on pre-crisis calm years only (2002–2006) and applied to all subsequent data. This prevents look-ahead leakage through feature normalization.

Evaluation. Leave-one-out cross-validation on the 7-sample annual training set (2002–2006 calm + 2008–2009 crisis). We report mean LOO AUC as the headline metric, not best-seed or full-sample AUC. Statistical significance is assessed via permutation test (1000 permutations of crisis labels).

2020 negative control. Models trained on the 2008 episode are applied to the 2020 episode without retraining. Theory predicts near-zero crisis probability for 2015–2019 (pre-COVID calm).

3 Results

3.1 Signal Analysis: Which Ecological Metrics Work?

Table 2 summarizes the pre-crisis behavior of all five ecological signals across both episodes. Figure 1 shows the signal trajectories.

Table 2: Ecological signal behavior across episodes. Kendall τ computed over pre-crisis years. For the 2020 negative control, theory predicts no significant trend (i.e., $|\tau|$ should be small and p large).

	2008 GFC (expect signal)		2020 COVID (expect no signal)	
Signal	τ	Verdict	τ	Verdict
Spectral gap	−0.87	✓ Declining	+0.20	✓ No trend
Housing starts var	+1.00	✓ Rising (peaks 2006)	−0.80	✓ Declining
EBI	+0.20	× Wrong direction	+0.80	∼ Rising
Housing starts AR(1)	−0.60	× Wrong direction	−0.40	✓ No buildup
TED spread var	+0.40	∼ Coincident only	+1.00	× Slow rise
Summary	2/5 signals as theorized		3/5 clean, 1 ambiguous, 1 FP	

fig1_signal_table.png

Figure 1: Ecological early warning signals, 2008 GFC episode. Red shading indicates crisis period (Dec 2007–Jun 2009). Verdicts indicate whether each signal moved in the theoretically predicted direction pre-crisis. Two of five signals behaved as theorized.

Spectral gap declined monotonically from 0.313 in 2002 to 0.221 in 2007 ($\tau = -0.87$), indicating progressive network fragmentation in the pre-crisis period. This is the cleanest ecological network signal.

Housing starts variance surged from 6,479 in 2002 to 25,323 in 2006 before declining—approximately 18 months before official crisis onset. This signal is methodologically clean: native monthly resolution, no interpolation.

EBI moved in the wrong direction. **Housing starts AR(1)** also went the wrong direction (becoming more negative, not rising toward 1). **TED spread variance** was elevated only at crisis onset—a coincident indicator, not a leading one.

3.2 The 2020 Negative Control

fig2_endogenous_vs_exogenous.png

Figure 2: Signal comparison: 2008 GFC (endogenous fold, signal expected) versus 2020 COVID (exogenous shock, no signal expected). The contrast is consistent with theoretical predictions for both working signals.

For spectral gap, the 2020 episode shows no directional trend pre-crisis ($\tau = +0.20$), compared to the monotonic decline before 2008 ($\tau = -0.87$). For housing starts variance, pre-COVID values were declining 2015–2019—the opposite of the pre-2008 buildup. These results are consistent with the theoretical prediction that CSD signals should be specific to endogenous bifurcation approaches.

3.3 False Positive Analysis

We test whether the spectral gap signal fires during calm periods (2010–2014 and 2015–2019). Spectral gap produces **zero false alarms** across ten calm-period years.

fig5_false_positive_analysis.png

Figure 3: False positive analysis. Spectral gap across all periods. Zero false alarms in ten calm-period years.

3.4 Trend Significance

fig6_kendall_tau.png

Figure 4: Kendall τ trend statistics across both episodes. Results are consistent with theoretical predictions for spectral gap and housing starts variance.

3.5 Keystone Sector Identification

Betweenness centrality analysis of the 2006 network using technical coefficient normalization identifies: Federal government enterprises (0.077), Chemical products (0.072), and **Federal Reserve banks, credit intermediation, and related activities (0.067)**. Finance correctly appears as the third most central sector, correcting the raw-flow specification failure where nominal volume effects caused Utilities and Chemicals to dominate.

fig4_network_structure.png

Figure 5: Network structure analysis. Left: keystone sectors, 2006. Right: spectral gap across episodes.

3.6 Predictive Performance

Table 3: Model performance. LOO cross-validation. RobustScaler fitted on pre-crisis years only. 2007 excluded from training labels.

Model	AUC	Evaluation	n
Logistic Regression (primary)	**1.000**	LOO CV	7
Random Forest	0.400	LOO CV	7
Permutation test (LR): $p = 0.034$, null mean = 0.483			
Advance warning test (2007): LR = 0.013, RF = 0.120			
2020 negative control: LR assigns 0.002–0.004 pre-crisis			
GNN (appendix only)	0.361 ± 0.440	5 seeds	7

Note: LOO AUC = 1.000 on n=7 has wide confidence intervals. The permutation test ($p = 0.034$) provides limited but non-trivial evidence. Multi-crisis validation is required. Random Forest underperforms, consistent with overfitting at small n.

fig7_results_summary.png

Figure 6: Honest claims and evidence summary.

Advance warning test. Neither model fires on 2007 (LR: 0.013, RF: 0.120). This is an honest null result.

2020 negative control (ML). Logistic regression assigns 0.002–0.004 crisis probability to 2015–2019 with zero false alarms—consistent with the theoretical prediction that ecological signals are quiet before an exogenous shock.

4 Discussion

4.1 Summary of Findings

1. Two of five ecological signals—spectral gap ($\tau = -0.87$) and housing starts variance (peaks 18 months pre-crisis)—behaved as theorized before the 2008 GFC.

2. The 2020 COVID negative control holds for four of five signals and for the ML

model: no pre-crisis ecological buildup before an exogenous shock.

3. Spectral gap produces zero false alarms across ten calm-period years.
4. Logistic regression achieves LOO AUC = 1.000 ($p = 0.034$); neither model provides advance warning in the year before crisis onset.
5. Three signals did not behave as theorized, providing honest evidence about the limits of the ecological mapping.

4.2 What the Results Do and Do Not Show

These results support the ecological framework as theoretically motivated and empirically promising for two specific signals. They do not demonstrate actionable real-time early warning, generalizability across crises, or incremental value beyond standard macroeconomic indicators.

The most meaningful result is the combination: spectral gap works before an endogenous crisis and is quiet before an exogenous one—precisely the discriminating pattern the theory predicts.

4.3 The Wrong Network Problem

The BEA IO network captures supply-chain flows, not financial exposure networks. The 2008 crisis propagated through mortgage securitization, interbank lending, and CDS exposures—none of which appear in IO tables. Applying ecological metrics to a financial exposure network from FFIEC Call Reports is the most important methodological improvement for future work.

4.4 The Theoretical Agenda

This paper computes the spectral gap of the adjacency matrix rather than the leading eigenvalue of the economic community (Jacobian) matrix—May's actual criterion. Properly computing May's criterion requires a dynamic Leontief-type model and a Jacobian with self-regulation terms on the diagonal. The leading eigenvalue of this Jacobian rising toward zero would unify the network fragmentation and CSD signals: both are manifestations of the same underlying quantity. This unification is the most important theoretical contribution available to this research program.

4.5 Limitations

Single positive episode. All metrics have wide confidence intervals. Multi-crisis leave-one-out validation is required.

Interpolation leakage. Cubic spline interpolation of annual BEA data uses future anchor points. Primary results use annual-resolution network metrics and native monthly CSD signals only.

Wrong network. IO network captures supply-chain flows, not financial contagion channels.

Signals that failed. EBI, housing AR(1), and TED spread variance did not behave as theorized. These failures constrain which ecological mechanisms are supported.

Robustness. Results are robust to crisis onset dating (AUC = 1.000 under both December 2007 and January 2008 onset specifications). Housing starts variance Kendall τ is positive across all window lengths tested (12–36 months), ranging from 0.60 to 1.00, and is statistically significant at $p < 0.05$ for windows of 24 months or longer. CSD signals alone achieve LOO AUC = 1.000; network-only achieves 0.400.

4.6 Future Work

1. **Multi-crisis panel.** Add 2001 episode and cross-country panels; evaluate with leave-one-crisis-out.

2. **Financial exposure network.** Construct from FFIEC Call Reports; apply EBI, spectral gap, and DebtRank.

3. **Economic community matrix.** Compute May's actual stability criterion; unify network and CSD signals under a single eigenvalue framework.

4. **Real-time data.** Pull ALFRED real-time vintages.

5. **Benchmark comparison.** Test incremental value beyond credit-to-GDP gap [9] and yield-curve inversion.

6. **GNN on actual sector graph.** 68-node IO graph with spatiotemporal architecture; pretrain on real IUCN extinction cascade data.

4.7 Conclusion

The ecological framework for systemic risk monitoring is theoretically motivated and produces two signals that behave as theorized. The 2020 negative control provides discriminating evidence: ecological signals are quiet before an exogenous shock and elevated before an endogenous one, consistent with fold bifurcation theory. Whether these signals provide actionable, generalizable early warning value beyond standard indicators remains open. We propose them as a theoretically grounded direction, with the methodological agenda above as the necessary next steps.

Acknowledgments

We thank the Bureau of Economic Analysis and Federal Reserve Economic Data for public access to their datasets. All code is available at `https://github.com/Abhiv1028/EcoEcon`.

A GNN Architecture and Full Results

The GNN is a preliminary methodology demonstration, not a primary result.

We construct a graph with 8 nodes (features) and edges connecting features with Pearson correlation $|r| > 0.3$, computed on pre-crisis training data only.

Table 4: GNN results (appendix). Not primary findings.

Model	AUC	Evaluation
GNN (no pretraining)	reported in notebook	LOO, 5 seeds
GNN (synthetic ecological pretraining)	reported in notebook	LOO, 5 seeds
Best-seed AUC = 0.993 selected by validation performance. Not an unbiased estimate. See notebook 03 for full results.		

The GNN uses the 8-node feature-correlation graph rather than the actual 68-node IO sector graph. Future work will use the actual sector graph with a spatiotemporal architecture and real IUCN extinction cascade pretraining.

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